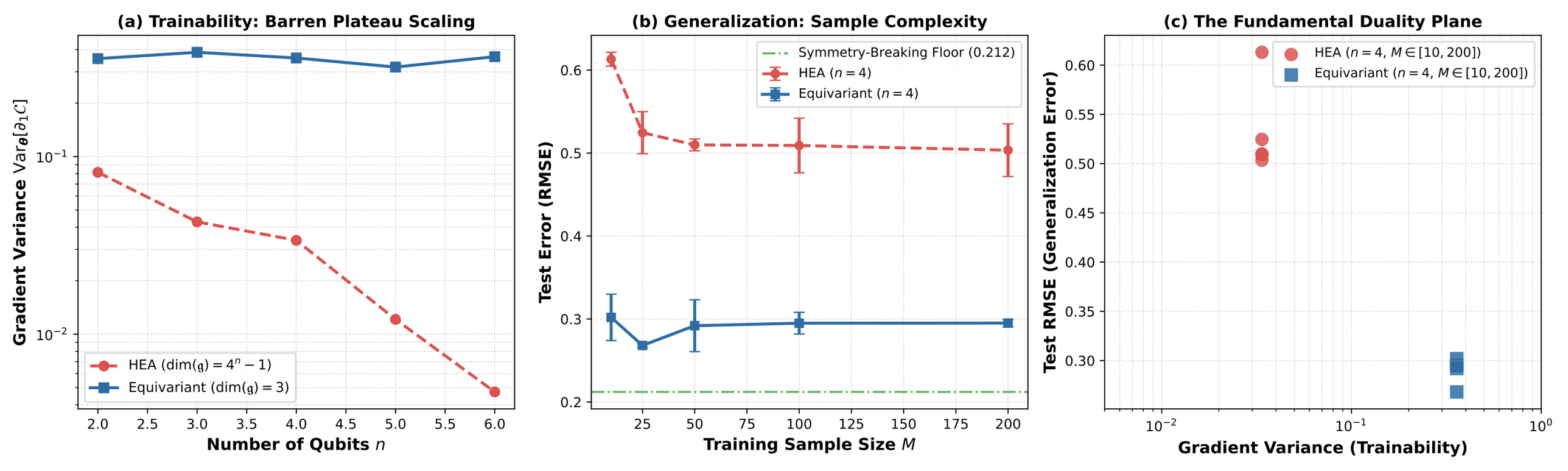
(a) Trainability: Barren Plateau Scaling
Gradient Variance $\mathrm{Var}_{\boldsymbol{\theta}}[\partial_1 \mathcal{C}]$
10^{-1}
10^{-2}
HEA ($\dim(\mathfrak{g}) = 4^n - 1$)
Equivariant ($\dim(\mathfrak{g}) = 3$)
2.0
2.5
3.0
3.5
4.0
4.5
5.0
5.5
6.0
Number of Qubits n
(b) Generalization: Sample Complexity
Test Error (RMSE)
0.6
0.5
0.4
0.3
0.2
Symmetry-Breaking Floor (0.212)
HEA ($n = 4$)
Equivariant ($n = 4$)
25
50
75
100
125
150
175
200
Training Sample Size M
(c) The Fundamental Duality Plane
Test RMSE (Generalization Error)
0.60
0.55
0.50
0.45
0.40
0.35
0.30
HEA ($n = 4$, $M \in [10, 200]$)
Equivariant ($n = 4$, $M \in [10, 200]$)
10^{-2}
10^{-1}
10^{0}
Gradient Variance (Trainability)